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Crab carapace shape as a biomarker of salt marsh metals pollution

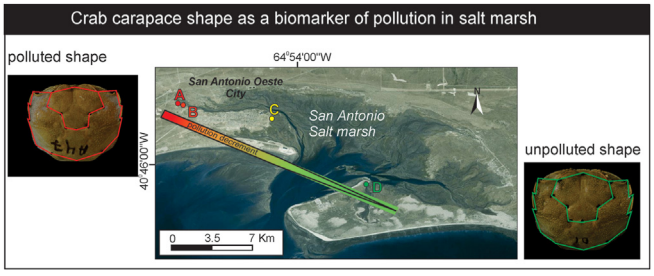
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HIGHLIGHTS

- Geometric morphometrics was used to study carapace shape variation in *N. granulata*.
- The symmetric component of the carapace shape variation responded to environmental stress.
- The asymmetric component was not associated with pollution.
- Carapace shape showed advantages with respect to other biomarkers.
- Carapace shape showed be a good biomarker of pollution stress in salt marsh crabs.

GRAPHICAL ABSTRACT



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ABSTRACT

Coastal environmental pollution is a global problem that has been growing for decades. For this reason, different approaches have been sought to address and detect its environmental implications. Some organisms are considered bioindicators or biomonitors of contamination, which provide information about environmental quality. Previous studies used the crab *Neoelice granulata* (Brachyura, Varunidae) as bioindicator of the presence of metals by the analysis of soft tissues to evaluate physiological and molecular markers. However, the contaminant levels accumulated in these soft tissues have shown to be affected by seasonal variations, suggesting that environmental assessments based on soft tissue samples may be unreliable. Within this framework, we aimed to describe the crab body (carapace) variations related to a known soil metal gradient in a Patagonian salt marsh and to evaluate the use of the body shape as an alternative biomarker for monitoring the quality of salt marsh systems. We studied the carapace shape variations using geometric morphometrics (GM) based on a 2D structure with object symmetry. We observed symmetric and asymmetric components of carapace shape variation. While the latter was not found associated with a gradient of contamination by metals, the symmetric component responded to environmental changes; therefore, it could be considered a stress biomarker related to metal contamination. Consequently, we recommended using GM analysis because it is inexpensive, faster and non-seasonal and could be used on living organisms, avoiding destroying individuals to measure the environmental stress.

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1. Introduction

One of the main anthropic threats is the production and

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discharge of dangerous substances into the marine environment (Williams, 1996). These are pollutants that generally affect the organisms' fitness of the intertidal habitats (Thompson et al., 2002; Weis, 2014). Metals are natural elements in the environment, however, their concentrations are increasing due to industrial progress (Rivera-Ingraham et al., 2011). The effect of exposure to metals (bioconcentration and bioaccumulation) are well documented on benthic marine organisms (Rainbow, 2002). Those effects have been widely examined in sentinel species (especially mollusks) to enable a more representative assessment of coastal pollution (Galloway et al., 2004). Biological, physiological and morphological responses (e.g. oxidative stress, DNA damage, metallothionein expression, histopathology, and variations in the form) have been commonly used to monitor exposure to and specific effects of hazardous substances on aquatic organisms as indicators of environmental health (ICES Advisory Committee on Marine Environment, 2003; Al-Subiai et al., 2011; Amira et al., 2018).

In recent years, the use of geometric morphometrics to study the shape variations in plants and marine invertebrates as indicators of pollution has increased (Ambo-Rappe et al., 2011; Márquez et al., 2011, 2015; Nuñez et al., 2012; Klisarić et al., 2014; Vujić et al., 2015; Smith, 2016; Pollicelli et al., 2018). When organisms are under environmental stress, the increase of fluctuating asymmetry (FA) has been interpreted as developmental instability of the symmetric structures (Møller and Shykoff, 1999). Several studies have however questioned the use of FA as a universal indicator of stress (Sandner et al., 2019). On the other hand, variations in the shape of structures without symmetry, as marine gastropods shells, were used as potential biomarkers of pollution (Márquez et al., 2011, 2017; Núñez et al., 2012; Primost et al., 2016; Harayashiki et al., 2020).

The salt marsh of San Antonio bay (Patagonia, Argentina) showed evidence of pollution due to the enrichment of metals in soils (Idaszkin et al., 2015; 2017a, b, c; Pollicelli et al., 2018), plants (Idaszkin et al., 2015, 2017a; Pollicelli et al., 2018) and marine invertebrate, such as crabs (Giarratano et al., 2016) and mussels (Vázquez et al., 2007). Previous studies determined a gradient in soil metal content; from a pile of remaining wastes of the closed Gonzalito mine to the open sea, within the salt marsh surrounding the San Antonio bay (Idaszkin et al., 2015; 2017a, b, c). This salt marsh is inhabited by cord grasses of the genus *Spartina* (*Spartina densiflora* Brongn and *Spartina alterniflora* Loisel.) and other halophytes such as *Sarcocornia perennis* (Mill.) A. J. Scott., *Limonium brasiliense* (Boiss.) Kuntze, and *Cressa truxillensis* Kunth. In salt marshes affected by pollution, soils and plants play an important role due to their ability to hold the pollutants, hence altering their dynamics in the environment (Hung and Chmura, 2007; Duarte et al., 2010; Almeida et al., 2011). One of the principal species structuring the salt marshes physiognomy is the semiterrestrial burrowing crab endemic to the warm temperate coasts of the Southwestern Atlantic, *Neoelice granulata* (Dana, 1851) (Brachyura, Varunidae) (Iribarne et al., 1997). This widely distributed species (from 23 °S Brazil to 41 °S Argentina) is a herbivorous detritivorous, consuming salt marsh plants (Luppi et al., 2013) and represents a fundamental link in the food web as prey (Giarratano et al., 2016). This species has been used as bioindicator of metal presence, using soft tissues, to evaluate physiological and molecular markers (Gil et al., 2006; Beltrame et al., 2011; Simonetti et al., 2013). However, according to Giarratano et al. (2016), contaminant levels accumulated in soft tissues of male crabs could be affected by seasonal variations, meaning that environmental assessments based on soft tissue samples may be unreliable. Therefore, the main goal of this study was to describe the crab body (carapace) variations related to a soil metal gradient and to evaluate the use the body shape as an alternative biomarker for monitoring the quality of salt marsh systems.

2. Material and methods

2.1. Sampling

We sampled within the salt marsh (Natural Protected Area of the Río Negro Province, Argentina, Fig. 1) located surrounding the San Antonio bay (40°44′S, 54°68′W). In January 2016, a total of 186 *Neohelice granulata* male crabs were collected and frozen (−20 °C) soon after collection, from four sites along the metals soil gradient as found by Idaszkin et al. (2015; 2017a, b, c). As Giarratano et al. (2016) suggested, depending on gender, crabs can show different patterns of metal accumulation and biochemical responses depending on gender. We choose to use only males to minimize variability and consider that only these have been used in previous works in the studied salt marsh (Giarratano et al., 2016). Besides, it is important to remark that using males as a biomarker presents less impact on reproductive populations.

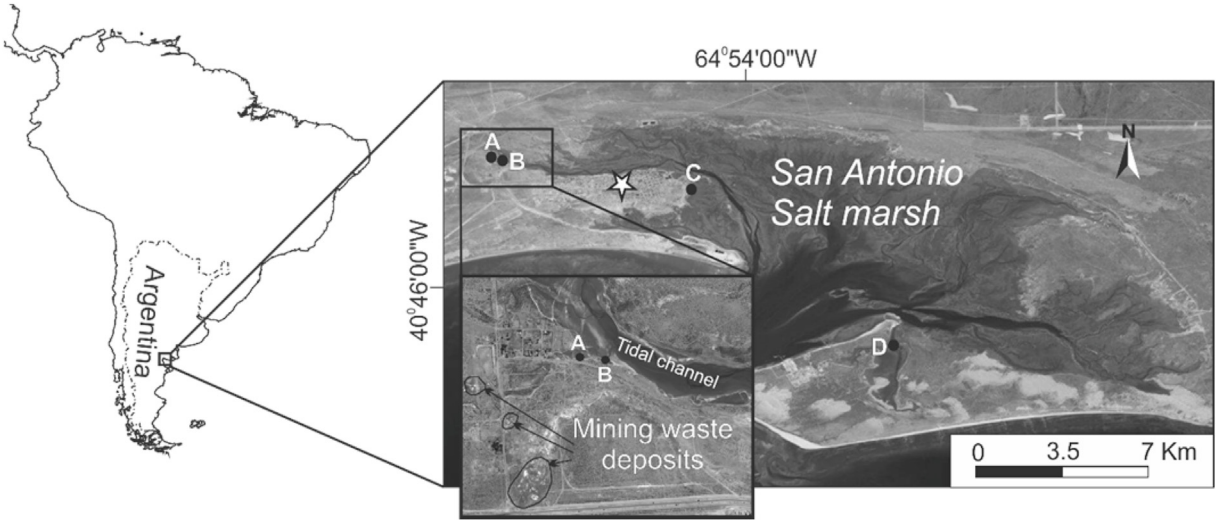


Fig. 1. Sample sites A, B, C and D from San Antonio salt marsh, Río Negro Province, Argentina. Star indicates the San Antonio city nearest to salt marsh.

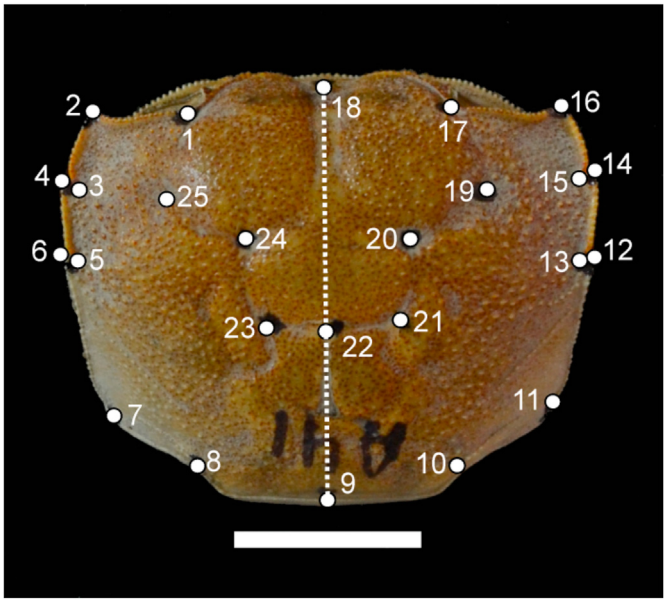


Fig. 2. Landmarks configurations on the *Neohelice granulata* used to represent the carapace shape variations in dorsal view. The landmarks are: (1–17) tip of the ocular depression, (2–16) external border of the ocular depression, (3–15) first lateral tooth, (4–14) external tip of the first lateral tooth, (5–13) second lateral tooth, (6–12) external tip of the second lateral tooth, (7–11) suture of the lateral side, (8–10) posterior base of the abdomen, (9) posterior edge of the axis of symmetry (defined as a vector between landmarks 18, 22 and 9, (18) anterior edge, (19–25) inflection lateral point of the hepatopancreas suture, (20–24) intersection between the cardiac suture and hepatopancreas suture, (21–23) distal point of the suture of the cardiac line, and (22) inflexion point on the cardiac suture.

Site A was in the higher topographic sector of the salt marsh, near the head of the main tidal channel and very close to the largest open-air dump receiving surface runoff from the mining drainage (Idaszkin et al., 2015). Site B was nearest site A in a topographically lower sector of the salt marsh. Site C was in the outer zone of the main tidal channel, in an external sector of the salt marsh and site D was outside the channel (Fig. 1) (Idaszkin et al., 2015). Site A showed high levels of Pb, Cu, and Zn in soils and plants, followed by sites B, C and D (Idaszkin et al., 2015; 2017a, b, c). All crabs were collected at low tide within the high marsh level, co-inhabited by the halophytes *Spartina densiflora* and *Sarcocornia perennis*.

Sex in male crabs was determined on the basis of the presence of the triangular shape of the abdominal perimeter. Carapace length (CL) was measured as the straight line between landmark 18 and 9 (Fig. 2). In total, 48 individuals from site A (CL: 1.9–2.7 cm), 48 from site B (CL: 1.8–2.8 cm), 44 from C (CL: 1.7–2.4 cm) and 46 from D (CL: 1.6–2.7) were used. Carapaces were dissected, carefully cleaned, and once dried, they were numbered and photographed in dorsal view.

2.2. Geometric morphometrics

Analyses of the carapace shape were performed using geometric morphometrics (GM) based on a 2D structure with object symmetry (Savriama and Klingenberg, 2011). The shape of the individuals was captured by Cartesian coordinates of 25 anatomical landmarks (Fig. 2). The carapace is symmetric regarding a perpendicular axis of bilateral symmetry (left-right axis) that was defined between landmarks 18, 22 and 9. All carapaces were digitized by one observer (FM) using the software TPSDig2 (Rohlf, 2017). To evaluate measurement error (ME), a subsample of 33 individuals was digitized twice. Since the carapaces presented symmetry, we made the Procrustes fit with reflection, first reflecting all landmark configurations from one side. Afterwards, a standard Procrustes fit was conducted (Klingenberg and McIntyre, 1998). After superimposition, the overall shape variation was preserved in the specimens aligned dataset, and variation around the mean shape (consensus) was separated into an asymmetry and symmetric component (Klingenberg et al., 2002).

2.3. Shape statistical analysis

The centroid size (CS) of individuals, defined as the square root of the sum of the squared deviations of landmarks from the centroid (Bookstein, 1991), was used as a proxy for shell size.

To study the presence of asymmetry and assess the relative magnitudes of ME, a Procrustes ANOVA (Klingenberg and McIntyre, 1998) test was conducted in MorphoJ, version 1.07a (Klingenberg, 2011). For Procrustes ANOVA, of the object symmetry, the units of analysis were left-right part of the carapace. The main effects were specimens (representing the inter-individual variation), sides (left-right representing the DA), specimens-by-side (representing the FA), and measurement error (replicate measurements) (Savriama and Klingenberg, 2011; Klingenberg, 2015). Also, the variable site was included in the model as an additional main effect (A, B, C and D). Specimens and carapace side were considered as random and fixed effects, respectively. Procrustes ANOVA also provided a simple means to gauge the possible effects of measurement error (ME) on estimates of FA (Klingenberg, 2015). The FA scores (computed using the Procrustes distances from Procrustes ANOVA) were used to measure the amount of asymmetry between right and left individual sides (Klingenberg and McIntyre, 1998; Klingenberg, 2015). We used one-way ANOVA test to evaluate statistical difference between FA site scores. Conversely, since the allometry can affect the shape variations of individuals, we estimated it through pooled-sites multivariate regression of Procrustes coordinates (dependent variables) on size (CS, independent variable). We described and illustrated the symmetry carapace shape variations using a Principal Component Analysis (PCA) of the symmetric component variance-covariance matrix. Finally, to find the shape components that maximize the separation among sites a Canonical Variate analysis (CVA) was performed, and the difference between Mahalanobis distances was tested with 10,000 permutations.

3. Results

Procrustes ANOVA test of shape variation revealed that all the effects of the model were statistically significant (Table 1). The sample presented both DA (main effect of side in Table 1) and FA (main effect of individual*side in Table 1). Measurement error (ME) was negligible in relation to individual*side effect since the mean square of FA largely exceeded ME (Klingenberg, 2015). We observed that Procrustes FA scores frequency histogram for the left–right differences were distributed around a mean that is not zero, indicating the presence of both DA and FA. We rejected the presence of antisymmetry, since no platykurtosis or binomial distributions were detected (data not shown). The Procrustes FA scores did not differ significantly among sites ($p = 0.58$). The relation between pooled within-sites size and symmetrical shape was significant ($p < 0.0001$) and accounted for 9.8% of the total shape variations. Therefore, subsequent shape analysis was done on the regression residuals, considered as allometry-free symmetrical shape variable.

The pattern of the symmetrical carapace shape variation is summarized in Fig. 3, showing the morpho-space defined of the first two principal components (PC). PC scores whose eigenvalues represented less of 10% of the total variations were not shown. Crabs located in the positive extreme of the PC1 (40.5%), which was

Table 1
Procrustes ANOVA results for carapace shape asymmetry. The table lists Procrustes mean squares (MS) and degrees of freedom (DF) for all effects, as well as parametric P-values.

Procrustes ANOVA				
Shape	MS	DF	F	p
Site	0.0003128422	69	8.14	<0.0001
Individual	0.0000384253	4186	3.19	<0.0001
Side	0.0004856029	23	40.31	<0.0001
Individual*Side	0.0000120470	4255	2.23	<0.0001
Measurement Error	0.0000054138	1472		

Values are significant at $p < 0.05$.

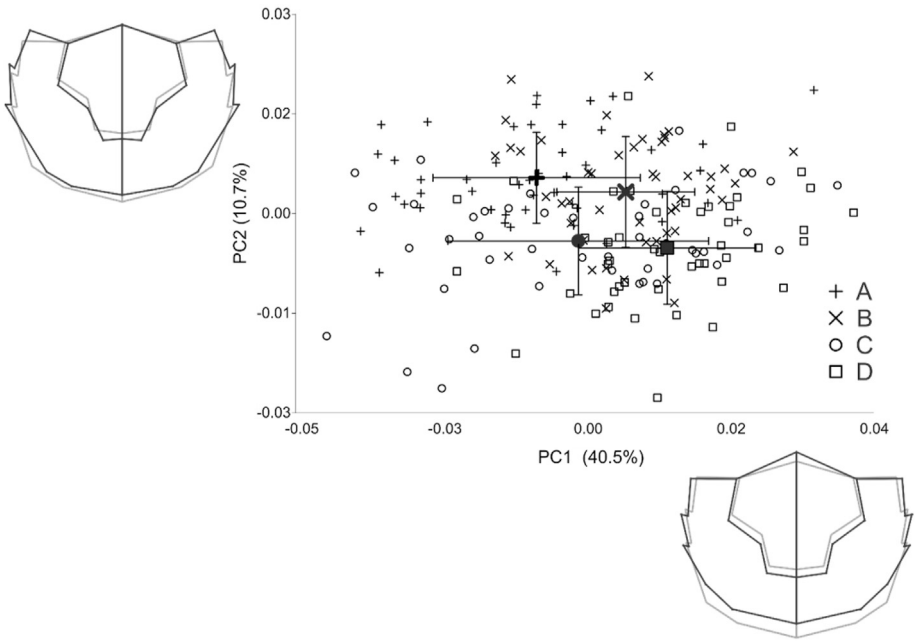


Fig. 3. Plot of the first two principal components (PC) showing the symmetrical carapace shape variation pattern. The wireframe shows the displacement vectors from consensus (grey wireframe) to the positive extreme shape (black wireframe) for each PC. Positive wireframe have a scale factor 0.1. The black largest symbols indicate the average for each group and the percentages of explained variance for each axis are in parentheses. See Fig. 1 for groups' reference labels.

associated with pentagon shape, presented an expansion in the rostral area, a constriction of the lateral and posterior part, with expanded of lateral teeth and cardiac and hepatopancreas sutures. In contrast, individuals toward the negative extreme showed speculate shell shape variations (more trapezoid shape). Individuals located in the positive extreme values of PC2 (10.7%) presented an anterior expansion of the external tips of the carapace (teeth and ocular area) and posterior extension of the hepatopancreas and cardiac sutures. On the contrary, individuals from the negative extreme showed the opposite shape variations (Fig. 3).

The among-sites symmetrical carapace shape variations were successfully discriminated using CVA. The first two CVs explicated 92.68% of the total shape component that maximized the site discrimination. The most significant differences in CVA were between the polluted (A and B) and unpolluted (C and D) sites. Unpolluted shapes were being associated with a more lateral and anterior expansion with a restricted posterior part of the carapace (Fig. 4). The among-sites comparison of mean carapace shape indicated that there were significant differences ($p < 0.0001$) among them (Fig. 4). All paired comparisons Mahalanobis distances among sites were statistically different. The Mahalanobis (shape) distances were shorter between the two polluted sites (A and B) and larger between A and D (unpolluted).

4. Discussion

Pollution studies can be approached throughout the direct measurement of pollutants in the environment (either in water, air or soil) or through the presence of pollutants in different living organisms. Some organisms that satisfy certain conditions (e.g., accumulate metals to high concentrations which may be relatively easily measured; are abundant at the sites of interest; easy to identify and sample; tolerant of exposure to environmental variations in physicochemical parameters (Rainbow and Phillips, 1993)) are considered bioindicators or biomonitors of contamination, which present the advantage of giving us information of environmental quality (Markert, 2007). Mollusks and crabs are widely used to evaluate toxicological impacts (Márquez et al., 2011; Simonetti at al., 2013; Lezcano et al., 2015; Primost et al., 2016). In particular, due to the fact that *N. granulata* shows tolerance to high levels of metals, accumulates them, and reflects the levels of contamination in the environment; it has been proposed as a potential bioindicator of pollution (Beltrame et al., 2011). However, in recent years, the use of biomarkers, which are measurable parameters at the sub-organismic (morphological, biological or physiological) whose changes indicate environmental effects, has gained prominence (Markert, 2007). Biomarkers are versatile (they can be measured in different organisms and environments) and are simple to measure.

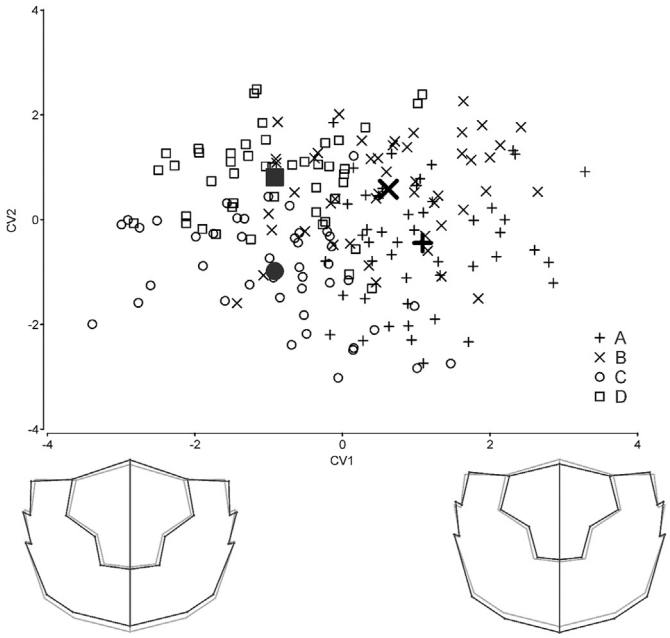


Fig. 4. First and second canonical axes showing the symmetrical shape component that most differentiate the four groups (A, B, C and D). The mean shape of each group is presented as the biggest symbol. Wire-frame diagrams show shape changes from the mean shape (grey vectors) to the positive (scale factor of 5) and negative extremes (scale factor of −5) on the canonical variate (CV1) axis. See Fig. 1 for groups' reference labels.

This paper evaluated the potential of a good, easy to measure, quantifiable stress biomarker (correspondence between the variation of the shape components (symmetric and asymmetric)) in a bioindicator organism, such as *N. granulata*, along a metal pollution gradient, as a promising tool to evaluate and monitor the quality of environments susceptible to the harmful effects produced by humans. Therefore, we evaluated the correspondence between the variation of the shape components (symmetric and asymmetric) associated with a gradient of metals pollution.

Fluctuating asymmetry (FA) is the capacity of the organisms to buffer themselves during the development stage as a response to different environmental stressors, such as pollutants (Klingenberg, 2015). The presence of FA was used, in general as an indicator of developmental instability. Lezcano et al. (2015) and Hayder-Ben Yahia and Selmi (2019) found elevated levels of FA in marine crabs, which they attribute to exposure to pollutants in carapace and legs, respectively. Our results were contrary in that they showed that FA was not related to polluted sites. Several other studies have corroborated these results in not detecting expected responses in FA to either genetic or environmental stressors (Clarke et al., 1992; Fowler and Whitlock, 1994; Vøllestad et al., 1998; Polak et al., 2002). Thus, the use of carapace FA as a biomarker should be done with restrictions, as it could be a species-specific response.

Among the metals, Pb is known to be harmful, and lacks biological function (Kumar et al., 2020). Giarratano et al. (2016) found a higher concentration of Pb in the soft tissues of *N. granulata* in the summer (the season of our sampling fieldwork), and a lower concentration of Pb and Cu in crabs from the sites far from the mine waste. However, these crabs could have a detoxification mechanism (Buzzi and Marcovecchio, 2016), so the metal concentrations in their tissues could underestimate the magnitude of the contamination and the state of health of the environment. Therefore, some biomarkers may not always be representative of the environmental contamination, and depend on the species in which they are measured. That is reflected by the lack of correlation between the environmental pollution reported for San Antonio bay, and enzymatic oxidative stress biomarkers measured in *N. granulata* by Giarratano et al. (2016). Likewise, the synthesis of metal sequestering proteins studied in *N. granulata* showed a direct relationship with the content of metals in their tissues, decreasing their concentration through the purification process (Buzzi and Marcovecchio, 2016). This purification process could occur in the crabs of the San Antonio salt marsh, in those seasons in which the metal concentrations decrease. The foregoing highlights the importance of looking for biomarkers with greater versatility and a high correspondence between their response and the pollution level. In this sense, Primost et al. (2016) and Gouveia et al. (2019) found a correspondence between the levels of variation in the shell shape of marine gastropods and a pollution gradient measured in the field. Regarding the symmetric shape analysis components, carapace shape (symmetrical components) was still observed between individuals collected in San Antonio salt marsh according to a pollution metals gradient. These results mirror the same pattern of the shape distances previously reported for *Cressa truxillensis* leaves and soil from the same sampled sites (Pollicelli et al., 2018). In agreement with this work, the individuals from polluted sites (A and B) presented different shapes to those collected from the unpolluted (C) ones.

On the base of the results found, we propose the use of the symmetrical shape variations rather than FA as a potential biomarker to evaluate the presence of metal pollutants in *N. granulata*. An advantage of the carapace shape analysis with respect to biochemical and molecular biomarkers is that the shape change is not affected due to seasonal variation. The biomarker presented here brings some promising insights concerning the evaluation and early monitoring of natural resources without destructive or extractive methods. It has a great utility to study carapace shape of organisms inhabiting protected areas or long-lived biomonitor species, with the advantage that it could be used on living organisms, thus avoiding the killing of individuals to measure environmental stress.

5. Conclusion

N. granulata shows a carapace shape variation, in both its symmetric and asymmetric components, but the latter was not found to be associated with a gradient of contamination by metals. The use of FA as an indicator of contamination is species-specific. However, the symmetric component responds to environmental changes that have already been determined in other groups of organisms such as halophytic plants and bivalves. Therefore, the symmetric component of the shape would be a stress biomarker associated with metal contamination.

Author contributions

Both authors contributed equal to this work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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